CLASSIFICATION

Algoritma Decision Tree

Key Problem

No	Savings	Assets	Income	Credit Risk
1	Medium	High	High	Good
2	Low	Low	Medium	Bad
3	High	Medium	Low	Bad
4	Medium	Medium	Medium	Good
5	Low	Medium	High	Good
6	High	High	Low	Good
7	Low	Low	Low	Bad
8	Medium	Medium	Medium	Good

Savings	Assets	Income	Credit Risk
Medium	Low	Medium	?
Low	High	High	?

Menghitung Impurity

Menghitung Kesamaan data (homogeneity) atau ketidaksamaan data (heterogeneity) dalam sebuah tabel yang mengandung atribut dan Kelas dari atribut.

Sebuah tabel dikatakan Pure atau Homogenous jika hanya mengandung satu class. Jika mengandung lebih dari satu kelas disebut Impure atau Heterogeneous.

Untuk menghitung Impurity dapat dilakukan dengan formula berikut :

$$Entropy = \sum_j -p_j log_2 p_j$$

$$Gini\ Index = 1 - \sum_j p_j^2$$

$$Classification\ Error = 1 - max\{p_j\}$$

Probability

	Atribut			Class
No	**Savings**	**Assets**	**Income**	**Credit Risk**
1	Medium	High	High	Good
2	Low	Low	Medium	Bad
3	High	Medium	Low	Bad
4	Medium	Medium	Medium	Good
5	Low	Medium	High	Good
6	High	High	Low	Good
7	Low	Low	Low	Bad
8	Medium	Medium	Medium	Good

Terdapat 3 Class Bad dan 5 Class Good. Total data 8 baris.

Probability Class adalah :

Prob (Bad) = 3 / 8 = 0.375
Prob (Good) = 5 / 8 = 0.625

Entropy Parent

$$Entropy = \sum_{j} -p_j log_2 p_j$$

Probability Class adalah :

Prob (Bad) = 3 / 8 = 0.375
Prob (Good) = 5 / 8 = 0.625

Entropy Parent = – 0.375 log (0.375) – 0.625 log (0.625)
= – 0.375 (– 0.426) – 0.625 (– 0.205)
= 0.15975 + 0.12815
= 0.29

Gini Index

$$Gini\ Index = 1 - \sum_j p_j^2$$

Probability Class adalah :

Prob (Bad) = 3 / 8 = 0.375
Prob (Good) = 5 / 8 = 0.625

Gini Index = $1 - (0.375^2 + 0.625^2)$
= $1 - (0.14 + 0.39)$
= $1 - 0.53$
= 0.47

Classification Error Index

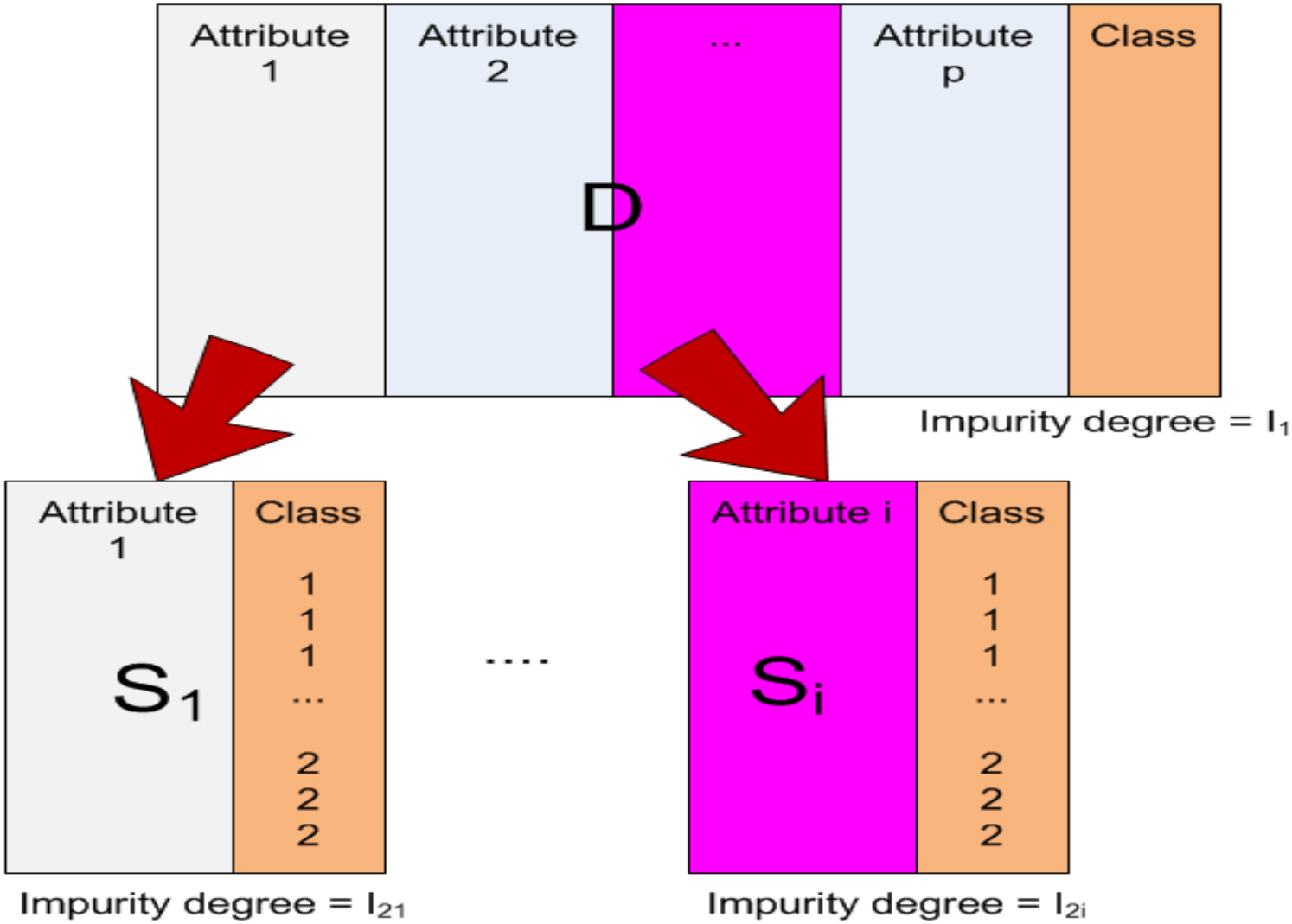
$$Classification\ Error = 1 - max\{p_j\}$$

Probability Class adalah :

Prob (Bad) = 3 / 8 = 0.375
Prob (Good) = 5 / 8 = 0.625

Classification Error Index = 1 – Max{0.375, 0.625}
= 1 - 0.625
= 0.375

Subset


Attribute 1
Attribute 2
...
Attribute p
Class
D
Impurity degree = I_1
Attribute 1
Class
1
1
1
...
2
2
2
S_1
....
Attribute i
Class
1
1
1
...
2
2
2
S_i
Impurity degree = I_{21}
Impurity degree = I_{2i}

Subset - Savings

	Atribut	Class
No	**Savings**	**Credit Risk**
1	Low	Bad
2	Low	Bad
3	Low	Good
4	Medium	Good
5	Medium	Good
6	Medium	Good
7	High	Bad
8	High	Good

No	Savings	Credit Risk
1	Low	Bad
2	Low	Bad
3	Low	Good

Prob (Bad) : 2/3 = 0.66
Prob (Good) : 1/3 = 0.33
Gini Index : $1 - (0.66^2 + 0.33^2)$
: 0.46

No	Savings	Credit Risk
4	Medium	Good
5	Medium	Good
6	Medium	Good

Prob (Bad) : 0/3 = 0
Prob (Good) : 3/3 = 1
Gini Index : $1 - (0^2 + 1^2)$
: 0

No	Savings	Credit Risk
7	High	Bad
8	High	Good

Prob (Bad) : 1/2 = 0.5
Prob (Good) : 1/2 = 0.5
Gini Index : $1 - (0.5^2 + 0.5^2)$
: 0.5

Subset - Assets

	Atribut	Class
No	**Assets**	**Credit Risk**
1	Low	Bad
2	Low	Bad
3	Medium	Bad
4	Medium	Good
5	Medium	Good
6	Medium	Good
7	High	Good
8	High	Good

No	Assets	Credit Risk
3	Medium	Bad
4	Medium	Good
5	Medium	Good
6	Medium	Good

Prob (Bad) : 1/4 = 0.25
Prob (Good) : 3/4 = 0.75
Gini Index : $1 - (0.25^2 + 0.75^2)$
: 0.375

No	Assets	Credit Risk
1	Low	Bad
2	Low	Bad

Prob (Bad) : 2/2 = 1
Prob (Good) : 0/2 = 0
Gini Index : $1 - (1^2 + 0^2) = 0$

No	Assets	Credit Risk
7	High	Good
8	High	Good

Prob (Bad) : 0/2 = 0
Prob (Good) : 2/2 = 1
Gini Index : $1 - (0^2 + 1^2) = 0$

Subset - Income

No	Income	Credit Risk
1	Low	Bad
2	Low	Bad
3	Low	Good
4	Medium	Bad
5	Medium	Good
6	Medium	Good
7	High	Good
8	High	Good

No	Income	Credit Risk
1	Low	Bad
2	Low	Bad
3	Low	Good

Prob (Bad) : 2/3 = 0.66
Prob (Good) : 1/3 = 0.33
Gini Index : $1 - (0.66^2 + 0.33^2)$
: 0.46

No	Income	Credit Risk
7	High	Good
8	High	Good

Prob (Bad) : 0/2 = 0
Prob (Good) : 2/2 = 1
Gini Index : $1 - (0^2 + 1^2) = 0$

No	Income	Credit Risk
4	Medium	Bad
5	Medium	Good
6	Medium	Good

Prob (Bad) : 1/3 = 0.33
Prob (Good) : 2/3 = 0.66
Gini Index : $1 - (0.33^2 + 0.66^2)$
: 0.46

Information Gain

Information Gain (i) Entropy :

Entropy dari Parent Tabel D – Sum ((Jumlah Data Subset / Jumlah Data Parent) * Entropy setiap Subset)

Information Gain (i) Gini Index :

Gini Index dari Parent Tabel D – Sum ((Jumlah Data Subset / Jumlah Data Parent) * Gini Index setiap Subset)

Information Gain (i) Classification Error :

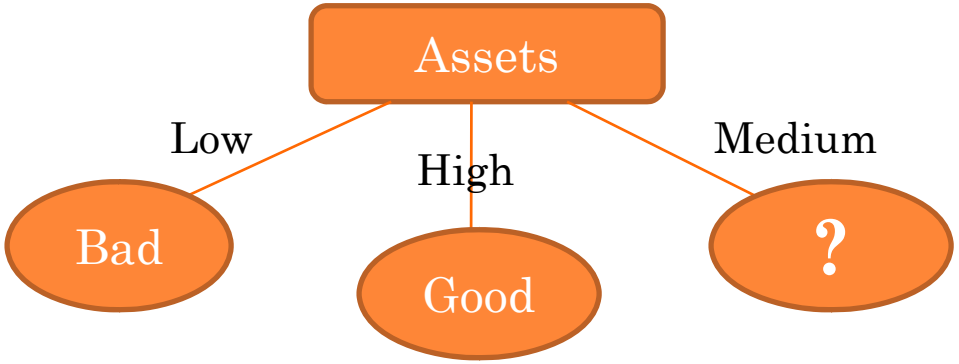
Classification Error dari Parent Tabel D – Sum ((Jumlah Data Subset / Jumlah Data Parent) Subset * Classification Error setiap Subset)

Information Gain

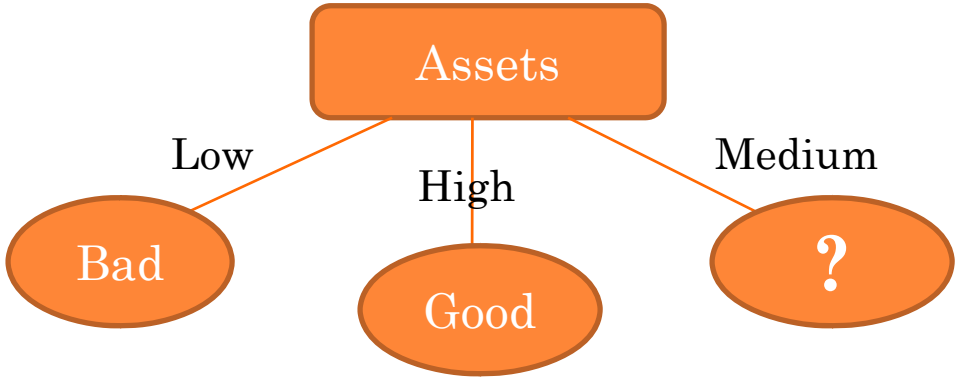
Gini Index Parent = 0.47, Jumlah Data Parent = 8

	Savings		Assets		Income	
Gini Index	Low (3)	0.46	Low (2)	0	Low (3)	0.46
	Medium (3)	0	Medium (4)	0.375	Medium (3)	0.46
	High (2)	0.5	High (2)	0	High (2)	0
Information Gain	0.47 – ((3/8 * 0.46) + (3/8 * 0) + (2/8 * 0.5)) **0.1725**		0.47 – ((2/8 * 0) + (4/8 * 0.375) + (2/8 * 0)) **0.2825**		0.47 – (3/8 * 0.46) + (3/8 * 0.46) + (2/8 * 0)) **0.125**	

Maximum Information Gain = Subset Assets
Pure (Homogen) Subset Assets = Low dan High



Decision Tree Rule



If Assets = Low Then Credit Risk = Bad
If Assets = High Then Credit Risk = Good

Parent - Iteration #2

No	Assets	Savings	Income	Credit Risk
1	Medium	High	Low	Bad
2	Medium	Medium	Medium	Good
3	Medium	Medium	Medium	Good
4	Medium	Low	High	Good

Prob (Bad) : 1/4 = 0.25
Prob (Good) : 3/4 = 0.75
Gini Index : $1 - (0.25^2 + 0.75^2)$
: 0.375

Subset Savings - #2

No	Savings	Credit Risk
1	High	Bad
2	Medium	Good
3	Medium	Good
4	Low	Good

No	Savings	Credit Risk
1	High	Bad

Prob (Bad) : 0/1 = 0
Prob (Good) : 1/1 = 1
Gini Index : $1 - (0^2 + 1^2) = 0$

No	Savings	Credit Risk
1	Medium	Good
2	Medium	Good

Prob (Bad) : 0/2 = 0
Prob (Good) : 2/2 = 1
Gini Index : $1 - (0^2 + 1^2) = 0$

No	Savings	Credit Risk
1	Low	Good

Prob (Bad) : 0/1 = 0
Prob (Good) : 1/1 = 1
Gini Index : $1 - (0^2 + 1^2) = 0$

Subset Income - #2

No	Income	Credit Risk
1	Low	Bad
2	Medium	Good
3	Medium	Good
4	High	Good

No	Income	Credit Risk
1	Low	Bad

Prob (Bad) : 1/1 = 0
Prob (Good) : 0/1 = 1
Gini Index : $1 - (1^2 + 0^2) = 0$

No	Income	Credit Risk
1	Medium	Good
2	Medium	Good

Prob (Bad) : 0/2 = 0
Prob (Good) : 1/1 = 1
Gini Index : $1 - (0^2 + 1^2) = 0$

No	Income	Credit Risk
1	High	Good

Prob (Bad) : 0/1 = 0
Prob (Good) : 1/1 = 1
Gini Index : $1 - (0^2 + 1^2) = 0$

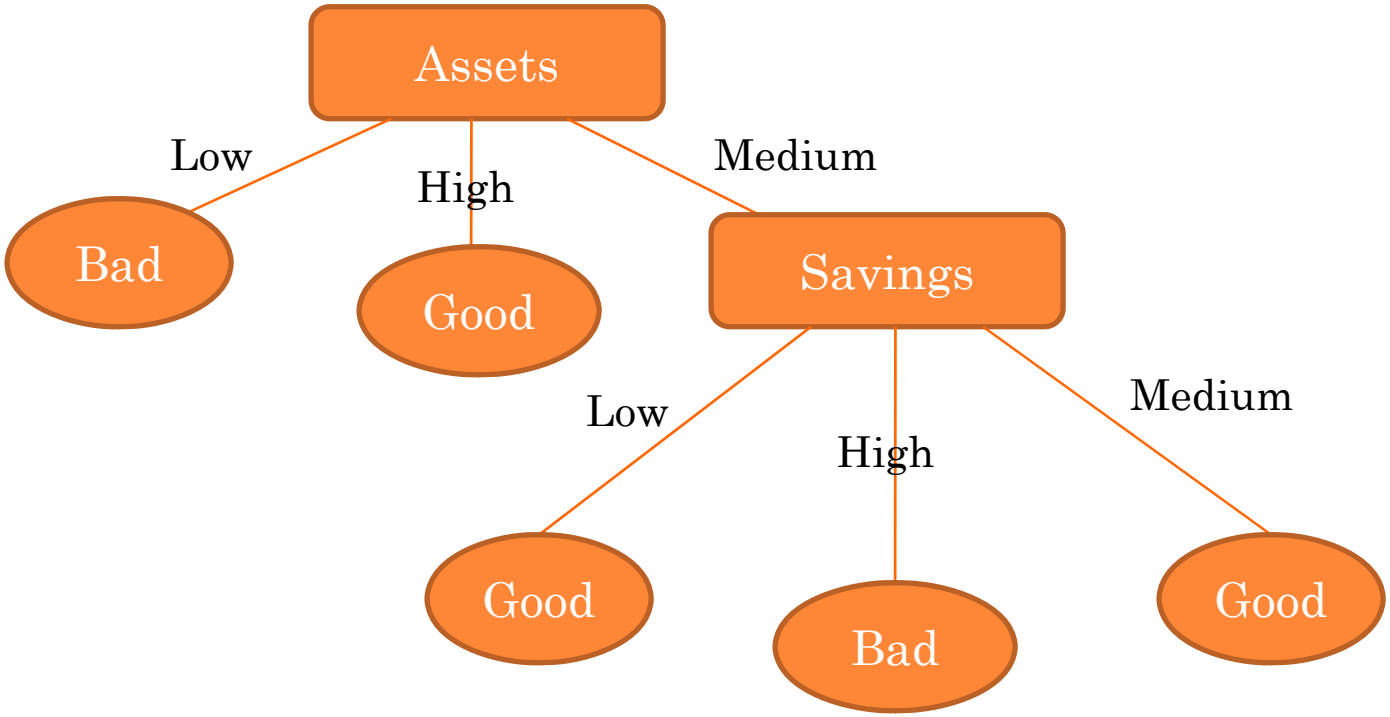
Information Gain - #2

Gini Index Parent = 0.375, Jumlah Data Parent = 4

	Savings		Income	
Gini Index	Low (1)	0	Low (1)	0
	Medium (2)	0	Medium (2)	0
	High (1)	0	High (1)	0
Information Gain	0. 375 – ((1/4 * 0) + (2/4 * 0) + (1/4 * 0)) **0.375**		0. 375 – ((1/4 * 0) + (2/4 * 0) + (1/4 * 0)) **0.375**	

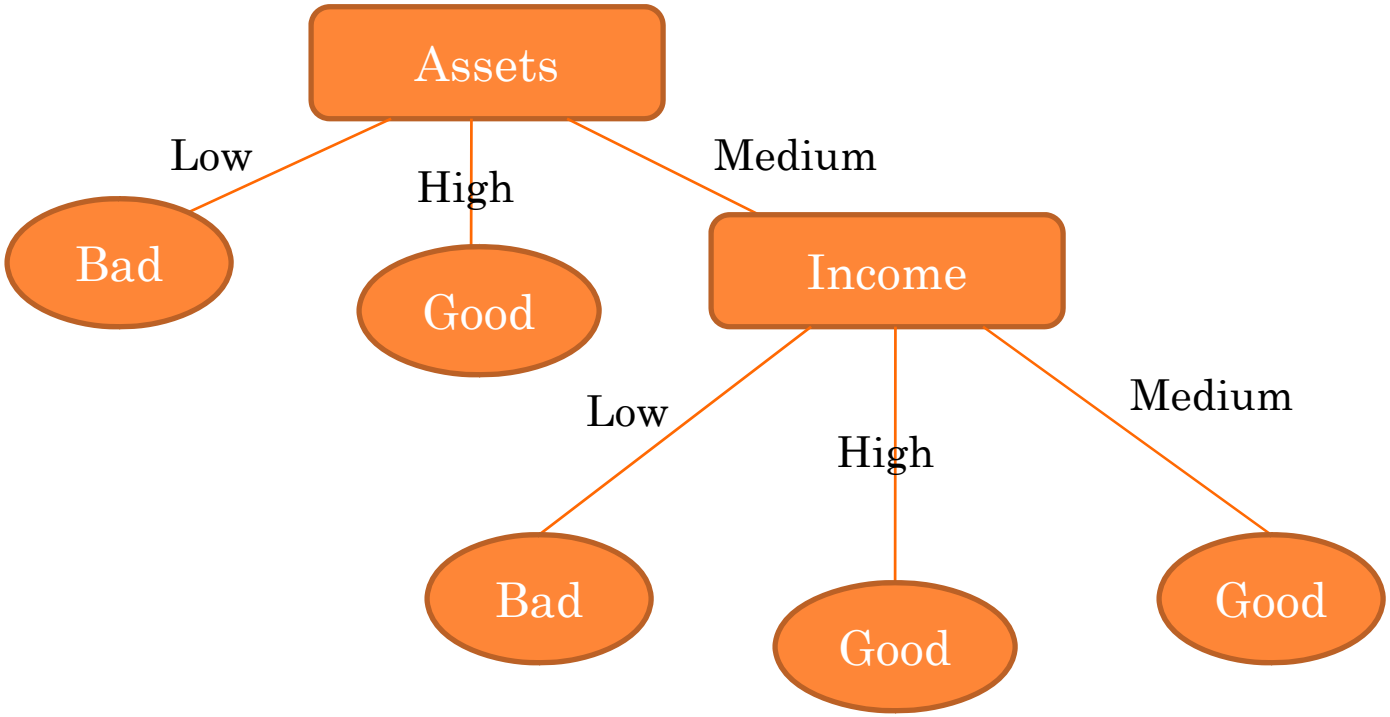
Maximum Information Gain = Subset Savings dan Subset Income
Pure (Homogen) Subset Savings = Low, Medium dan High
Pure (Homogen) Subset Income = Low, Medium dan High

Decision Tree Rule #2 Versi Savings



#1. If Assets = Low Then Credit Risk = Bad
#2. If Assets = High Then Credit Risk = Good
#3. If Assets = Medium And Savings = Low Then Credit Risk = Good
#4. If Assets = Medium And Savings = High Then Credit Risk = Bad
#5. If Assets = Medium And Savings = Medium Then Credit Risk = Good

Decision Tree Rule #2 Versi Income



#1. If Assets = Low Then Credit Risk = Bad
#2. If Assets = High Then Credit Risk = Good
#3. If Assets = Medium And Income = Low Then Credit Risk = Bad
#4. If Assets = Medium And Income = High Then Credit Risk = Good
#5. If Assets = Medium And Income = Medium Then Credit Risk = Good

Decision Tree Rule - Result

#1. If Assets = Low Then Credit Risk = Bad
#2. If Assets = High Then Credit Risk = Good

#3a. If Assets = Medium And Savings = Low Then Credit Risk = Good
#4a. If Assets = Medium And Savings = High Then Credit Risk = Bad
#5a. If Assets = Medium And Savings = Medium Then Credit Risk = Good

#3b. If Assets = Medium And Income = Low Then Credit Risk = Bad
#4b. If Assets = Medium And Income = High Then Credit Risk = Good
#5b. If Assets = Medium And Income = Medium Then Credit Risk = Good

Savings	Assets	Income	Credit Risk	Savings Or Income
Medium	Low	Medium	?	Bad / Bad
Low	High	High	?	Good / Good

References

- Discovering Knowledge in Data (Introduction to Data Mining), Chapter 6, Daniel T. Larose, Wiley, 2004

Attributes				Classes
Gender	**Car Ownership**	**Travel Cost ($)/km**	**Income Level**	**Transportation Mode**
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Male	0	Expensive	Medium	

Name	Give Birth	Lay Eggs	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	no	yes	mammals
python	no	yes	no	no	no	reptiles
salmon	no	yes	no	yes	no	fishes
whale	yes	no	no	yes	no	mammals
frog	no	yes	no	sometimes	yes	amphibians
komodo	no	yes	no	no	yes	reptiles
bat	yes	no	yes	no	yes	mammals
pigeon	no	yes	yes	no	yes	birds
cat	yes	no	no	no	yes	mammals
leopard shark	yes	no	no	yes	no	fishes
turtle	no	yes	no	sometimes	yes	reptiles
penguin	no	yes	no	sometimes	yes	birds
porcupine	yes	no	no	no	yes	mammals
eel	no	yes	no	yes	no	fishes
salamander	no	yes	no	sometimes	yes	amphibians
gila monster	no	yes	no	no	yes	reptiles
platypus	no	yes	no	no	yes	mammals
owl	no	yes	yes	no	yes	birds
dolphin	yes	no	no	yes	no	mammals
eagle	no	yes	yes	no	yes	birds